LoRA Meets Similarity Distribution Matching: Parameter-Efficient CLIP Tuning for Text-Based Person Search

Đỗ Thanh Tùng
Student ID: 20252432M
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***Abstract*—Text-based person search (TBPS) retrieves images of a pedestrian from a gallery given a free-form description. Surveillance data are often held by operators who cannot pool them, which makes federated learning (FL) attractive, but every FL client uploads its trainable parameters in every round. A federated TBPS system built on a large backbone therefore needs a much smaller trainable part that costs little accuracy. We test this premise centrally on IRRA, the CVPR 2023 method studied in this course, which fine-tunes all of CLIP with the Similarity Distribution Matching (SDM) loss. Freezing CLIP and adding rank-4 LoRA to the attention projections of both encoders, we train 491 520 parameters instead of 149.6 M, 304× fewer. With SDM as the only loss the model reaches 58.85% Rank-1 on RSTPReid at its best epoch and 57.18% after convergence, on par with the 57.20% IRRA reports for full fine-tuning with SDM alone; for 15 clients, the upload per round would drop from 8.4 GiB to 28.1 MiB. We then ask why SDM beats the standard contrastive loss. Under random batch sampling only 1.52% of batch rows contain a second pair of the same person, so SDM's target is almost always one-hot. In that regime SDM is not InfoNCE but a scaled mean-absolute-error loss plus an entropy bonus. A control loss with SDM's targets and the opposite KL direction stays within 0.18 Rank-1 points of InfoNCE on average and 3.15 points below SDM, which places the gain in the direction of the divergence rather than in multi-positive supervision.**

***Index Terms*—text-based person search, cross-modal retrieval, CLIP, LoRA, parameter-efficient fine-tuning, federated learning, contrastive loss**

I. Introduction

Text-based person search (TBPS) ranks a gallery of pedestrian images against a natural-language query such as "a woman in a white coat carrying a black backpack" [1]. It is cross-modal retrieval of the kind multimedia databases serve: the gallery is indexed once by an image encoder, each query is embedded by a text encoder, and retrieval is a nearest-neighbour search in the shared space. The difficulty is granularity. Every gallery item is a person, and a description has to separate people who differ only in the colour of a garment or the bag they carry.

Vision–language models pre-trained on web-scale image–text pairs, CLIP in particular [2], are now the backbone of the strongest TBPS methods [3], [4]. IRRA [5], the CVPR 2023 paper studied in this course, fine-tunes the whole CLIP ViT-B/16 with three objectives: implicit relation reasoning (IRR), identity (ID) classification and Similarity Distribution Matching (SDM). Its ablation on RSTPReid [6] shows that SDM contributes the most of the three. Parameter-efficient transfer learning (PETL) methods for TBPS [7]–[9] have since shown that a frozen backbone can compete, but they still train millions of parameters spread over several modules, which makes the effect of the loss hard to isolate.

Why parameter efficiency matters here. Surveillance footage belongs to operators who often cannot share it, so a TBPS model may have to be trained where the data reside, with federated learning (FL) [10], [11]. FL pays for the trainable part again and again: every client downloads and uploads it in every round, for tens of rounds, often over slow links [11]. Fully fine-tuning CLIP would mean 571 MiB per client per round. Before non-IID clients and aggregation even come into play, a federated design with parameter-efficient fine-tuning (PEFT) needs one thing to be true: a much smaller trainable part must cost little accuracy. This report checks that premise in the centralised setting, where it can be measured cleanly, as groundwork for federated TBPS with PEFT. Federated experiments are outside its scope.

We study the leanest configuration we could build: a frozen CLIP backbone, low-rank adaptation (LoRA) [12] of the attention projections, and SDM as the only loss. Two questions guide the report. **(Q1)** How much accuracy does this configuration give up relative to IRRA's fully fine-tuned counterpart, and what does it save? **(Q2)** Why does SDM outperform CLIP's InfoNCE loss [13]? IRRA attributes the gain to SDM counting every same-identity pair in a batch as a positive, and we test that explanation directly. The answer also matters for FL. SDM is the only IRRA objective without parameters tied to the global identity set, and whether its gain survives on clients that hold few images per person depends on why it helps.

Existing work and our contributions. Table I lists what this report takes from prior work and what it adds, separately for the model, the theory and the experiments. The section headings follow the same split: Section III [Existing] only summarises CLIP and IRRA, while Sections IV and V [Ours] are our own work and cite each borrowed element where it is used. Our contributions are:

1) A LoRA+SDM pipeline for TBPS that trains 0.33% of CLIP's parameters and reaches 58.85% (peak) and

TABLE I
WHAT THIS REPORT TAKES FROM PRIOR WORK AND WHAT IT ADDS.

Existing (source)	Ours
Model	
CLIP ViT-B/16 dual encoder [2]; 384×128 input, τ=0.02 [5]; LoRA [12]	Frozen backbone with LoRA on the Q/K/V/O projections of *both* encoders; choice of rank, scaling and schedule
Theory	
InfoNCE [13]; SDM and its multi-positive rationale [5]; MAE robustness [14]; confidence penalty [15]	In-batch positive rate (Eq. (6)); SDM with one-hot targets as scaled MAE plus entropy bonus (Eq. (7)); its gradient (Eq. (8)); MPCE as a control (Eq. (9))
Experiments	
RSTPReid and its protocol [6]; IRRA's full fine-tuning results, quoted and not rerun [5]	All LoRA runs; InfoNCE/MPCE/SDM ablation; parameter, storage and communication accounting

57.18% (converged) Rank-1 on RSTPReid, against 57.20% for IRRA's full fine-tuning with SDM alone (Section V-B), with an estimate of the savings in storage, optimiser memory and federated communication (Section V-D).
2) An analysis of SDM under identity labels: the rate of second positives in a random batch (none in 98.48% of rows), a closed-form decomposition showing that SDM with one-hot targets is not InfoNCE, and its gradient (Section IV-B).
3) A controlled loss ablation under LoRA, in which a multi-positive cross-entropy (MPCE) control separates the effect of identity-aware targets from that of the KL direction (Section V-C).

II. Related Work

Text-based person search. Li *et al.* [1] introduced the task with CUHK-PEDES; ICFG-PEDES [16] and RSTPReid [6], both built from MSMT17 imagery, followed. Early methods paired separately pre-trained image and text networks and aligned them with identity-aware objectives, most notably the cross-modal projection matching and classification losses CMPM/CMPC [17], often with explicit part- or phrase-level alignment [16]. CLIP changed the starting point. CFine [3] and the empirical study of Cao *et al.* [4] show that a jointly pre-trained image–text model outperforms these specialised designs, and IRRA [5] combines full CLIP fine-tuning with masked-language-model relation reasoning and the SDM loss. We keep IRRA's backbone and loss and drop everything else.

Parameter-efficient transfer. PETL freezes a pre-trained model and trains a small inserted component: bottleneck adapters [18], learned prefixes [19], or LoRA [12], which adds a trainable low-rank update to frozen weight matrices that can be merged into them after training. For TBPS, CSKT [7] transfers CLIP knowledge with prompts and adapters, DM-Adapter [8] routes features through a mixture of adapters, and UP-Person [9] unifies LoRA, prefixes and adapters. Each trains several million parameters across several modules and losses. We use one LoRA configuration an order of magnitude smaller and one loss, so the role of the loss can be measured.

Federated TBPS. FedAvg [10] trains a shared model by averaging client updates, so the size of the trainable part sets the communication cost of each round. FedSH [11], the first federated TBPS framework, uses ResNet-50 and Bi-LSTM backbones and exchanges only part of the model to cut that cost; it also notes that two-tower models double the transmission per round. We are not aware of federated TBPS work that pairs a CLIP backbone with PEFT. Section V-D estimates what LoRA would save in that setting.

III. Background: CLIP and IRRA [Existing]

This section summarises CLIP [2] and IRRA [5] and fixes the notation used later. None of it is our work.

Dual encoder. An image I_i is encoded by a ViT-B/16 vision transformer [20] and a description T_j by a 12-layer text transformer; the outputs are projected to a shared 512-dimensional space and ℓ_2-normalised, giving $\mathbf{v}_i$ and $\mathbf{t}_j$. Following IRRA, images are resized to 384×128, which preserves a standing person's aspect ratio; the pre-trained 14×14 grid of position embeddings is resampled bicubically to the resulting 24×8 patch grid. Retrieval ranks gallery images by the cosine similarity $\mathbf{v}_i^\top \mathbf{t}_j$.

Batch distributions. For a mini-batch of B image–text pairs with identity labels $y_1, \dots, y_B$, the image-to-text matching distribution of row i is

$$p_{i,j} = \frac{\exp(\mathbf{v}_i^\top \mathbf{t}_j/\tau)}{\sum_{k=1}^{B} \exp(\mathbf{v}_i^\top \mathbf{t}_k/\tau)}, \qquad \tau = 0.02, \tag{1}$$

and the text-to-image distribution is defined symmetrically. CLIP's InfoNCE loss treats only the paired caption as correct:

$$\mathcal{L}_{\text{nce}} = \tfrac{1}{2}\big(\mathcal{L}^{\text{i2t}} + \mathcal{L}^{\text{t2i}}\big), \qquad \mathcal{L}^{\text{i2t}} = -\frac{1}{B}\sum_i \log p_{i,i}. \tag{2}$$

SDM. IRRA argues that this target is wrong whenever another pair in the batch shows the same person, and replaces it with the identity-aware distribution

$$q_{i,j} = \frac{\mathbf{1}[y_i = y_j]}{\sum_k \mathbf{1}[y_i = y_k]}. \tag{3}$$

SDM then minimises $\mathrm{KL}(p_{i,\cdot} \| q_{i,\cdot})$, with the *predicted* distribution in the first argument and a small $\epsilon = 10^{-8}$ that keeps the logarithm finite:

$$\mathcal{L}^{\text{i2t}}_{\text{sdm}} = \frac{1}{B}\sum_i \sum_j p_{i,j} \log \frac{p_{i,j}}{q_{i,j} + \epsilon}, \quad \mathcal{L}_{\text{sdm}} = \mathcal{L}^{\text{i2t}}_{\text{sdm}} + \mathcal{L}^{\text{t2i}}_{\text{sdm}}. \tag{4}$$

IRRA's other components. IRR adds a cross-modal interaction encoder (13.66 M parameters [5]) that predicts masked words from the image, and the ID loss adds a linear classifier over all training identities. IRRA's ablation on RSTPReid (its Table 4, reproduced in our Table II) reports 54.05% Rank-1 for InfoNCE alone, 57.20% with SDM alone, and 60.20% with all three objectives. IRRA trains for 60 epochs with Adam at a learning rate of 10^{-5} and draws training batches uniformly at random.

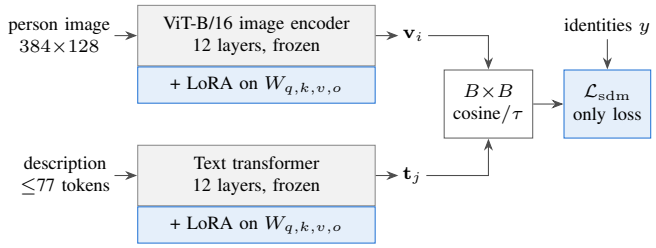


Fig. 1. The pipeline. Grey: frozen CLIP ViT-B/16 (149.6 M parameters). Blue: the only trainable parameters, 96 LoRA modules of rank 4 (491 520 parameters), trained with SDM alone.

IV. Method and Analysis [Ours]

A. LoRA-tuned CLIP

Following LoRA [12], we replace every linear projection inside multi-head attention (the query, key, value and output maps $W \in \mathbb{R}^{d\times d}$) by

$$W'\mathbf{x} = W\mathbf{x} + \frac{\alpha}{r} BA\,\mathbf{x}, \qquad A \in \mathbb{R}^{r\times d},\ B \in \mathbb{R}^{d\times r}, \tag{5}$$

with W frozen, $r = 4$ and $\alpha = 8$. A is initialised Kaiming-uniform and B to zero, so training starts exactly from pre-trained CLIP. Applied to both encoders, this gives $12\times4\times2 = 96$ modules (Fig. 1). Each module holds $r(d+d)$ parameters: the image encoder ($d = 768$) contributes $48{\cdot}4{\cdot}1536 = 294{,}912$ and the text encoder ($d = 512$) $48 \cdot 4 \cdot 1024 = 196{,}608$, a total of 491 520, or 0.33% of the 150.1 M parameters of the adapted model. Everything else stays frozen, including the embeddings, layer norms, feed-forward blocks, projection heads and the temperature. Because BA can be added to W after training, inference costs the same as CLIP.

Objective. We train with $\mathcal{L}_{\text{sdm}}$ of Eq. (4) and nothing else. Besides isolating the loss, this keeps the trainable part small. IRRA's ID classifier alone would add 512×3701 weights plus biases, 1.90 M parameters or nearly four times our LoRA budget, and the IRR encoder another 13.66 M. SDM has no parameters and uses the labels only through the relation $y_i = y_j$.

B. What SDM computes with identity labels

How often does a batch contain a second positive? SDM's target differs from InfoNCE's only in rows of q with more than one non-zero entry. RSTPReid's training split has 37 010 image–caption examples, ten per identity (five images, two captions each). With uniform random batches of $B = 64$, as used by IRRA and in our experiments, the probability that a row has another example of its identity in the batch is

$$1 - \prod_{k=0}^{B-2} \frac{N-10-k}{N-1-k} = 1.52\%, \qquad N = 37{,}010. \tag{6}$$

In 98.48% of rows, therefore, $q_{i,\cdot}$ is the one-hot vector $\mathbf{e}_i$, exactly InfoNCE's target. Whatever separates SDM from InfoNCE in these rows has to come from the form of the loss.

SDM with a one-hot target is not InfoNCE. Substituting $q_{i,\cdot} = \mathbf{e}_i$ into one row of Eq. (4) and writing $\lambda = \log(1/\epsilon)$ gives

$$\begin{aligned}\ell_i^{\text{sdm}} &= \sum_j p_{i,j}\log p_{i,j} + \lambda \sum_{j\neq i} p_{i,j} - p_{i,i}\log(1+\epsilon) \\ &\approx \underbrace{\lambda\,(1-p_{i,i})}_{\text{scaled MAE}} - \underbrace{H(p_{i,\cdot})}_{\text{entropy bonus}},\end{aligned} \tag{7}$$

with $\lambda = 18.42$, whereas InfoNCE's row loss is $\ell_i^{\text{nce}} = -\log p_{i,i}$. The first term is the mean absolute error between $p_{i,\cdot}$ and $\mathbf{e}_i$, $\frac{1}{2}\|p_{i,\cdot} - \mathbf{e}_i\|_1 = 1 - p_{i,i}$, scaled by λ. MAE is bounded and symmetric, and is known to be robust to label noise for that reason [14]. The second term rewards high-entropy predictions. It is the confidence penalty of Pereyra *et al.* [15], here weighted $1/\lambda \approx 0.05$ relative to the first. Both losses are minimised by $p_{i,\cdot} = \mathbf{e}_i$, but they weight the examples very differently on the way there.

Gradients. For logits $z_{i,j} = \mathbf{v}_i^\top \mathbf{t}_j/\tau$ and $\tilde{q} = q + \epsilon$, the exact gradient of one SDM row is

$$\frac{\partial \ell_i^{\text{sdm}}}{\partial z_{i,j}} = p_{i,j}\big(\log p_{i,j} - \log \tilde{q}_{i,j} - \ell_i^{\text{sdm}}\big), \tag{8}$$

while InfoNCE gives $p_{i,j} - \mathbf{1}[j = i]$. On the matching pair, InfoNCE pushes with strength $1 - p_{i,i}$, largest for the pairs the model ranks worst. The dominant MAE term of Eq. (7) pushes with $\lambda\, p_{i,i}(1 - p_{i,i})$, which vanishes as $p_{i,i} \to 0$. SDM therefore concentrates on pairs the model already half-recognises and largely ignores those it cannot match, the reverse of InfoNCE. In TBPS such pairs are often ambiguous rather than informative: two people in near-identical clothing, or a caption that omits the one distinguishing detail.

A control loss that isolates the KL direction. To test whether this, and not the identity-aware target, explains SDM's advantage, we add a control loss, multi-positive cross-entropy:

$$\begin{aligned}\mathcal{L}_{\text{mpce}}^{\text{i2t}} &= -\frac{1}{B}\sum_i \sum_j q_{i,j}\log p_{i,j} \\ &= \frac{1}{B}\sum_i \Big[\,\mathrm{KL}(q_{i,\cdot}\|p_{i,\cdot}) + H(q_{i,\cdot})\Big],\end{aligned} \tag{9}$$

summed over both directions as for SDM. MPCE uses SDM's targets with the *forward* KL; $H(q)$ does not depend on the model. The three losses then give two controlled contrasts: MPCE versus InfoNCE changes only the target (identity-aware or diagonal), and SDM versus MPCE changes only the KL direction. With one-hot targets MPCE equals InfoNCE up to the factor 2 of Eq. (2), and Adam-type optimisers are invariant to a constant rescaling of the loss (up to their ϵ and gradient clipping). If our explanation is right, MPCE should behave like InfoNCE and not like SDM.

V. Experiments [Ours]

A. Setup

Data and metrics. RSTPReid [6] contains 20,505 images of 4,101 identities with two descriptions per image, split into 3,701 training, 200 validation and 200 test identities. The test

protocol ranks a gallery of 1,000 images against 2,000 textual queries. We report Rank-K accuracy (R@K, the fraction of queries whose first correct image appears in the top K), mean average precision (mAP) and mean inverse negative penalty (mINP) [5]. With 2,000 queries, R@1 is a binomial estimate whose standard error is about 1.1 points near 58%; mAP averages over complete rankings and is less noisy.

Training details. We start from the publicly released CLIP ViT-B/16 weights [2]. Training images are flipped, padded, randomly cropped and randomly erased, and captions are truncated to 77 tokens. We train with AdamW [21] (peak learning rate 10^{-4}, weight decay 0.02), batch size 64, gradient clipping at norm 5 and mixed precision. The learning rate rises linearly over 3 warm-up epochs and then follows a cosine decay to 1% of its peak; one epoch takes about 7.2 minutes on a single GPU. The SDM run uses a 20-epoch schedule and the InfoNCE and MPCE runs a 30-epoch schedule, InfoNCE being stopped after 27 epochs. Warm-up is identical, and up to epoch 9 the learning rates stay close (at epoch 4, 9.92 vs. 9.97×10^{-5}; at epoch 9, 7.26 vs. 8.84×10^{-5}).

Protocol. We evaluate after every epoch. Like IRRA, we report the epoch with the best R@1 (*peak*). Selecting an epoch on the test split is optimistic, so we also report the *converged* value, the mean $\pm$ standard deviation over the last six epochs.

B. LoRA versus full fine-tuning (Q1)

Table II answers Q1. The fair comparison is IRRA's ablation row that, like ours, uses SDM alone. Full fine-tuning reaches 57.20% R@1; LoRA reaches 58.85% at its peak and 57.18$\pm$0.28% after convergence. The peak is 1.65 points higher, about 1.5 standard errors, and the converged value matches, so with a single seed we read the two as on par rather than claim an improvement. LoRA gets there with 0.33% of the parameters and 1.88 MiB of trained weights instead of 571 MiB. The complete IRRA model, which adds the ID loss and the IRR encoder, is ahead by 1.35 points of R@1 and 1.71 of mAP at the peak. Closing that gap would take extra objectives with their own parameters, the trade-off made by the PETL methods in Section II.

C. Where does SDM's gain come from? (Q2)

Table III and Fig. 2 compare the three losses under identical LoRA training. SDM leads InfoNCE by 3.65 points of R@1 and 3.31 of mAP at the peak, close to the 3.15 points IRRA reports for full fine-tuning, so LoRA keeps SDM's benefit. The ablation also shows where that benefit comes from.

Identity-aware targets contribute nothing measurable. MPCE and InfoNCE share the same 27-epoch schedule, so they can be compared epoch by epoch. Across these epochs MPCE's R@1 differs from InfoNCE's by 0.18$\pm$0.59 points on average (largest gap 1.45) and its mAP by 0.17 (largest 0.64), both well inside run-to-run noise. The ratio of their training losses stays between 2.03 and 2.12, as Section IV-B predicts: exactly 2 for one-hot rows, slightly more because of the 1.52% of rows with an extra positive, which confirms that the targets are almost always one-hot.

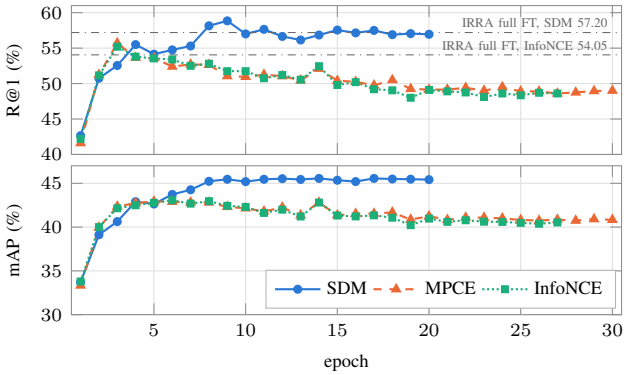


Fig. 2. Test R@1 and mAP after each epoch. MPCE and InfoNCE peak at the end of warm-up and then decline together; SDM keeps improving. Grey lines: IRRA's full fine-tuning results [5].

The KL direction accounts for the whole gain. SDM and MPCE share the target and differ only in the direction of the divergence, and SDM leads MPCE by 3.15 points of R@1 and 3.11 of mAP at the peak. The curves show how this gap arises. InfoNCE and MPCE peak at epoch 3, the end of warm-up, and then lose 6.60 and 6.70 points of R@1 while their training losses keep falling (InfoNCE from 2.10 to 0.29), the usual sign of overfitting. SDM instead climbs until epoch 9 and ends within 1.90 points of its peak. Over epochs 4–9 the learning rates of the two schedules differ by at most 18%, yet the curves move in opposite directions (SDM from 55.50 to 58.85, InfoNCE from 53.80 to 51.75), so the schedule difference cannot explain the gap.

Interpretation. Under random sampling, IRRA's rationale for SDM, that it accounts for several positives per batch, applies to fewer than 2% of rows and cannot explain a three-point gain. Equations (7) and (8) offer an explanation consistent with the curves: InfoNCE spends the largest gradients on the pairs it ranks worst, which on a small dataset with ambiguous descriptions leads it to memorise them, whereas SDM's bounded, MAE-like term down-weights those pairs and its entropy bonus discourages over-confident similarity distributions. We have not separated the two terms experimentally, so this remains a hypothesis, but the decomposition makes it testable: varying ϵ rescales λ and with it the weight of the MAE term against the entropy bonus, and each term can be trained alone.

D. Implications for federated training

Table IV turns the parameter count into the quantities that dominate FL, assuming that every client takes part in every round and uploads its whole trainable part. With full fine-tuning each round would send 8.4 GiB upstream, a heavy load for camera-side clients, whereas a whole LoRA round sends 20$\times$ less than a single client's full-model upload. Each client's optimiser state shrinks by the same factor, which matters on edge hardware. Together with Section V-B, this is the premise a federated PEFT design for TBPS needs: the 304$\times$ reduction costs no measurable accuracy against SDM-only full fine-

TABLE II

RESULTS ON RSTPREID (TEXT-TO-IMAGE, %). ROWS MARKED IRRA ARE QUOTED FROM [5] (TABLES 3 AND 4 THERE; THAT ABLATION REPORTS NO MAP OR MINP). OUR ROWS ARE SINGLE RUNS AT THEIR PEAK EPOCH; CONVERGED VALUES ARE IN TABLE III. "STORED" IS THE SIZE OF THE TRAINED PARAMETERS IN FP32.

	Method	Tuning	Loss	Trainable	Stored	R@1	R@5	R@10	mAP	mINP
IRRA	CLIP baseline	full	InfoNCE	149.6 M*	571 MiB	54.05	80.70	88.00	43.41	22.31
	SDM only (Tab. 4, No. 2)	full	SDM	149.6 M*	571 MiB	57.20	79.90	88.10	–	–
	IRRA (full model)	full	SDM+ID+IRR	>165 M†	>630 MiB	**60.20**	**81.30**	88.20	**47.17**	**25.28**
Ours	LoRA r=4 (ep. 3)	LoRA	InfoNCE	0.49 M	1.88 MiB	55.20	76.65	85.25	42.15	20.64
	LoRA r=4 (ep. 3)	LoRA	MPCE	0.49 M	1.88 MiB	55.70	76.40	85.50	42.35	20.78
	LoRA r=4 (ep. 9)	LoRA	SDM	0.49 M	1.88 MiB	58.85	80.30	**89.15**	45.46	23.29

*CLIP ViT-B/16 with position embeddings resampled to the 24×8 grid (149 617 665 parameters). †CLIP plus the 13.66 M IRR encoder [5] and the 1.90 M ID classifier; the MLM head is not counted.

TABLE III

LOSS ABLATION WITH LORA r=4 (SINGLE SEED, %). ONLY THE LOSS DIFFERS. CONVERGED VALUES AVERAGE THE LAST SIX EPOCHS OF EACH RUN'S OWN SCHEDULE.

Loss	Target	Divergence	Peak R@1 (ep.)	Peak mAP	Conv. R@1
InfoNCE	diagonal	CE	55.20 (3)	42.15	48.52±0.25
MPCE	identity	$\mathrm{KL}(q\|p)$	55.70 (3)	42.35	48.85±0.15
SDM	identity	$\mathrm{KL}(p\|q)$	**58.85** (9)	**45.46**	**57.18**±0.28

TABLE IV

WHAT THE TRAINABLE-PART SIZE IMPLIES FOR A FEDERATION OF K=15 CLIENTS (ONE PER RSTPREID CAMERA) OVER R=30 ROUNDS. ARITHMETIC FROM THE PARAMETER COUNTS (FP32), NOT A MEASUREMENT; DOWNLOADS FROM THE SERVER DOUBLE THE TRAFFIC.

	Full FT	LoRA r=4	Ratio
Trainable parameters	149.6 M	0.49 M	304×
Upload per client per round	571 MiB	1.88 MiB	304×
Upload per round (K clients)	8.4 GiB	28.1 MiB	304×
Upload over R rounds	251 GiB	0.82 GiB	304×
AdamW state per client	1.11 GiB	3.75 MiB	304×
R@1, SDM only (peak / conv.)	57.20 / –	58.85 / 57.18	

tuning and 1.35 points of R@1 against the complete IRRA model.

The loss analysis carries over too. Unlike IRRA's ID classifier, whose rows refer to identities that other clients may never see, SDM has no parameters tied to the global identity set, so it can be optimised locally and averaged without modification. And since SDM's gain does not depend on several images of one person sharing a batch (Section V-C), there is no reason to expect it to vanish on clients with few images per identity. Whether these arguments hold under non-IID clients and the averaging of low-rank factors can only be settled by a federated experiment, which is beyond this report.

VI. LIMITATIONS

Every run uses one seed, and R@1 on RSTPReid has a standard error of about 1.1 points, so differences below roughly two points, such as LoRA versus full fine-tuning with SDM, should be read as parity. The SDM-versus-InfoNCE and SDM-versus-MPCE gaps are about three standard errors and also appear in the less noisy mAP. The SDM run used a shorter learning-rate schedule than the other two; Section V-C argues that this does not explain the result, but a matched rerun would remove the doubt. All conclusions concern RSTPReid with random batch sampling. On larger datasets, or with an identity-balanced sampler that puts several images of each person in a batch, the multi-positive targets of SDM and MPCE become frequent and may then matter. The IRRA numbers are quoted from the paper, not reproduced, and the mechanism proposed in Section V-C has not been isolated experimentally. Finally, no federated experiment is reported: Section V-D is an estimate from parameter counts, and the behaviour of accuracy under non-IID clients remains to be measured.

VII. CONCLUSION

This report tested, in the centralised setting, the premise behind PEFT for federated text-based person search: that a far smaller trainable part costs little accuracy. With 0.33% of the parameters, 304× fewer than full fine-tuning, LoRA+SDM matches IRRA's fully fine-tuned SDM-only model on RSTPReid (58.85% peak and 57.18% converged, against 57.20%) and trails the complete IRRA model by 1.35 points of R@1. For a 15-client federation this would cut the upload per round from 8.4 GiB to 28.1 MiB, so the premise holds in the centralised setting. On the second question our evidence contradicts the usual explanation. Under random sampling SDM's targets are one-hot in 98.48% of rows, where SDM acts as a scaled mean-absolute-error loss plus an entropy bonus, and a control loss with SDM's targets but the forward KL behaves like InfoNCE. SDM's benefit therefore comes from the direction of its divergence, which resists the overfitting InfoNCE shows on this small dataset. Next steps are seeded reruns with a common schedule, ablations of the two terms of Eq. (7), an identity-balanced sampler, and federated training of LoRA+SDM over camera-level clients on RSTPReid, CUHK-PEDES and ICFG-PEDES.

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